

ATISHAY JAIN

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EDUCATION

Brown University

September 2021 - May 2026 (Expected)

PhD, Computer Science | Focus: Machine Learning and Foundation Models

GPA: 4/4

IIT (Indian Institute of Technology), Gandhinagar

August 2016 - July 2020

B.Tech, Computer Science and Engineering | President's Gold Medalist

GPA: 10/10

SKILLS

Languages: Python, R, C, Java, C++

Libraries: Pytorch, HuggingFace, Keras, TensorFlow, Sklearn, CUDA, PyTorch Geometric, Spacy, NLTK

Soft Skills: Cross-Functional Collaboration, Project Orchestration, Technical Communication, Adaptability

Generative & Foundational AI: Zero-shot enhancement of Foundation Models, GANs, Auto-encoders, Diffusion frameworks, MLOps, Environment Management, Reproducibility, Full-stack ML Pipeline Development

Model Efficiency and Interpretability: Large-scale inference, memory optimization, Resource-efficiency, Explainers, SHAP, Attention

Sequence & Graph Modeling: NLP, GNNs, LSTMs, RNNs, Streaming data algorithms (Sketching), Audio processing

GRADUATE RESEARCH PROJECTS (ADVISOR: DR. RITAMBHARA SINGH)

Zero-Shot Enhancement of Foundation Models via Graph Diffusion

- Designed a graph diffusion framework to integrate spatial information in spatial transcriptomics data to improve the performance of existing foundation models with no extra training
- Conducted extensive comparative analysis against multiple foundation models and demonstrated a 30% to 130% performance improvement, even in comparison to specialized methods
- [Submitted](#) to RECOMB 2026

Nested Graph Neural Networks for Improved Spatial Clustering

- Implemented a nested Graph Neural Network (GNN), designing a custom architecture for unsupervised clustering of spatial transcriptomics data to enable cell annotation and understand disease pathology
- Validated model performance through rigorous benchmarking against state-of-the-art techniques, GraphST and MuCoST, achieving a 12% to 17% average performance gain
- [Published](#) peer-reviewed research at ACM-BCB 2025 and presented posters at Intelligent Systems for Molecular Biology (ISMB), Montreal 2024 and Cold Spring Harbor Laboratory, 2024 New York, demonstrating strong technical communication and documentation skills

Resource-Efficient Semantic Segmentation via Graph Neural Networks

- Developed a graph neural network-based framework to perform memory-efficient semantic segmentation on large microscopy images
- Reduced computational overhead by 10x–1000x (1–3 orders of magnitude) compared to standard U-Nets, while maintaining competitive pixel-level accuracy benchmarks
- [Published](#) in Microscopy journal and presented at Cold Spring Harbor Laboratory, 2022 New York

Comparative Evaluation Framework for Data-Upscaling Architectures

- Surveyed and reviewed multiple state-of-the-art models that upscale Hi-C data to reduce sequencing costs
- Conducted rigorous performance testing across heterogeneous datasets using multiple quantitative metrics to identify advantages and disadvantages of the models and methods used to train them
- [Published](#) findings in a peer-reviewed publication in the Genes journal

WORK EXPERIENCE (INTERNSHIPS)

Caltech (California Institute of Technology); Dr. Mani Chandy; [May 2019 - July 2019]

- Engineered Python algorithms for Caltech's IoTpy framework to process heterogeneous streaming data (sensors, network logs, audio, web-clicks)
- Developed heavy hitter algorithms and an online version of PCA for real-time analysis

IIT Gandhinagar; Dr. Anand Sengupta; [May 2018 - December 2019]

- Built ML de-noising models to identify binary black hole mergers in gravitational wave data
- Analysed neural network models, which use RNNs, LSTMs with attention, Auto-Encoders, and GANs to rapidly de-noise gravitational waves of varying signal-to-noise ratios

Hewlett Packard Enterprise; Mr. Suhas Shivanna; [June 2017 - July 2017]

- Designed a Generic Data Mining Domain Modeller for network telemetry data

SELECTED PROJECTS

Cross-Modal Alignment for Unpaired & Heterogeneous Data; Dr. Randall Balestriero

- Architected a novel multimodal pretraining strategy integrating ViTs and BERT to align heterogeneous unpaired data (images, text, clinical logs) using Gromov-Wasserstein Optimal Transport
- Implemented distributed training pipelines (DDP) on GPU clusters and optimized composite loss functions, achieving superior performance on multiple benchmarks with limited supervision compared to CLIP baselines

Defending Neural Networks Against Adversarial Attacks; Dr. Nipun Batra

- Explored and adapted state-of-the-art defenses such as ensembles, defensive distillation, and Defense-GAN against adversarial attacks

Detecting Insults in Social Commentary; Dr. Mayank Singh

- Used text pre-processing (NLP), and ML classifiers such as logistic regression, neural networks and SVMs for offensive comment identification on social media

PUBLICATIONS

- **Jain A.**, Pham, T.M., Laidlaw D.H., Ma Y., and Singh R. Diffusion-based Representation Integration for Foundation Models Improves Spatial Transcriptomics Analysis. Submitted to *Research in Computational Molecular Biology (RECOMB)*, May 2026 [[conference_link](#)]
- **Jain A.**, Laidlaw D.H., Ma Y., and Singh R. Improved Spatial Transcriptomics Clustering with Nested Graph Neural Networks. *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB)*, October 2025 [[link](#)][[conference_link](#)]
- **Jain A.**, Laidlaw D.H., Bajcsy P., and Singh R. Memory-efficient semantic segmentation of large microscopy images using graph-based neural networks. *Microscopy*, Oct 2023 [[link](#)] [[journal_link](#)]
- Murtaza G., **Jain A.**, Hughes M., Wagner J., and Singh R. A Comprehensive Evaluation of Generalizability of Deep Learning-Based Hi-C Resolution Improvement Methods. *Genes*, Dec 2023 [[link](#)] [[journal_link](#)]
- Gohil V.*, Narayanan S.D.*, and **Jain A***. [Re] One ticket to win them all: generalizing lottery ticket initializations across datasets and optimizers. *ReScience C 6, 2, #4*, Feb 2020 [[link](#)] [[journal_link](#)]
- Dutta R.*, Gohil V.*, and **Jain A***. Effect of Feature Hashing on Fair Classification. *ACM India Joint International Conference on Data Science and Management of Data (CoDS-COMAD)*, Hyderabad, India, Jan 2020 [[link](#)] [[conference_link](#)]

INVITED TALKS AND POSTERS

Talk at ACM-BCB 2025, Philadelphia - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Lightning Talk and Poster at CSHL Biological Data Science Meeting 2024, New York - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Poster at Intelligent Systems for Molecular Biology 2024, Montreal - Clustering Spatial Transcriptomics Data with Nested Graph Neural Networks

Talk at CSHL Biological Data Science Meeting 2022, New York - Scalable and memory efficient segmentation of large microscopy images using graph-based neural networks

Talk at PyData Gandhinagar 2018 - Applications of Machine Learning Across Domains

HONORS AND RECOGNITION

- Awarded the [President's Gold Medal](#) for obtaining the highest GPA among all B.Tech students graduating in 2020 from IIT Gandhinagar.
- Awarded the [Scholarship for Academic Excellence](#) for obtaining first rank in class, IIT Gandhinagar for the years 2018-19, 2019-20
- Appeared on the [Dean's list](#) (higher than 9.0 Term GPA) for all semesters at IIT Gandhinagar
- Placed in the top 0.2% of IIT-JEE (Joint Entrance Examination), 2016 with over 1.1 million candidates
- Won [HP Code Wars](#), Bangalore, 2015 - a prestigious high-school coding competition conducted by Hewlett Packard where approximately 100 teams participated

LEADERSHIP & PROFESSIONAL SERVICE

- Sub-reviewer for Nature, ICML, NeurIPS, ICLR, and RECOMB conferences.
- Co-founded [Gandhinagar chapter of PyData](#) and organised 4 meetups to expand membership over 1000
- Mentored over 300 students as a teaching assistant across courses related to Python and deep learning.